International Journal of Information Systems and Supply Chain Management
Volume 19 • Issue 1 • January-December 2026

Enhancing Resilience in Logistics:
The Role of HR-SCM Practices and AI Technology Readiness

Yibin Li
Chang'an University, China

Aicha Ndiaye
https://orcid.org/0009-0001-3067-6787
Chang'an University, China
Accepted: December 17th, 2025

ABSTRACT

This paper examines the role of human resources in supply chain management practices in fostering organizational resilience and transformative resilience within logistics firms, with a focus on the moderating effect of artificial intelligence technology readiness. Drawing on dynamic capabilities theory and supply chain resilience frameworks, the study proposes a model in which cross-training, managerial support, and flexible shift design act as key enablers of resilience. Using a cross-sectional survey, data were collected from 280 employees across various logistics companies in China to explore the relationships among human resources in supply chain management practices, organizational resilience, and transformative resilience. This study contributes to the growing body of transformative supply chain research by examining the connections between workforce adaptability, digital readiness, and resilience outcomes. The results suggest that organizational resilience, supported by robust human resource practices and artificial intelligence integration, is essential for sustaining long-term resilience and competitive advantage amid global disruptions.

KEYWORDS

Organizational Resilience, Transformative Resilience, HR Practices, Cross-Training, Managerial Support, Dynamic Capabilities Theory, Artificial Intelligence Readiness

INTRODUCTION

Recent global disruptions, such as the COVID-19 pandemic, have highlighted the vulnerability of supply chains while underscoring their crucial role in societal functioning. With issues, such as climate change, political unrest, and shifting customer expectations, these events have significantly reshaped the global supply chain landscape (He et al., 2023; Wieland, 2021). The growing recognition of these challenges underscores the need for academic research to go beyond traditional firm performance metrics and address the broader role of supply chains in supporting societal resilience (Mollenkopf et al., 2024). To remain relevant, supply chain management (SCM) research must evolve to address complex societal problems, known as "wicked problems," that require innovative, long-term solutions (Pederneiras et al., 2022).

Studies show that goals and expectations of the next generation of SCM professionals are shifting, with a greater focus on ethical and social outcomes (Dawood & Afzal, 2025). This generational

DOI: 10.4018/IJISSCM.401244

This article published as an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creative-commons.org/licenses/by/4.0/) which permits unrestricted use, distribution, and production in any medium, provided the author of the original work and original publication source are properly credited.

shift requires research to address complex challenges that extend beyond operational efficiency to encompass organizational resilience and artificial intelligence (AI) technology readiness, both of which are crucial for maintaining supply chain continuity during crises (He et al., 2023; Liang et al., 2025). Our study focuses on organizational resilience as a key underlying mechanism linking human resource management within supply chain frameworks (HR-SCM) practices (such as cross-training, flexible shifts, and managerial support) to transformative resilience, particularly in the logistics industry. These shifts reflect the changing expectations of both workers and organizations, as well as the evolving role of digital transformation in enabling organizations to manage and thrive through disruption (He et al., 2023).

There is a gap in societal issues, including racial justice, worker safety, and customer satisfaction, that has become more pronounced in supply chain decision-making. The highly publicized 2013 United Parcel Service strike focused attention on driver welfare, underscoring the importance of worker well-being in last-mile logistics (Esper et al., 2010). Furthermore, issues, such as unsafe working conditions and ethical labor practices, continue to challenge organizations, particularly in the context of global supply chains (Wieland & Wallenburg, 2013). Given these challenges, supply chains must shift toward transformative practices that promote societal well-being while ensuring business continuity (Wieland, 2021). As such, the academic community must increasingly incorporate frameworks, such as transformative supply chain research (TSCR), which views supply chains as enhancing both organizational and societal resilience (Mirzabeiki & Aitken, 2023; Mollenkopf et al., 2024).

This study makes several significant theoretical contributions. It advances the field by embedding workforce microfoundations, such as cross-training and adaptive routines, within the frameworks of both dynamic capabilities (Teece, 2007b) and supply-chain resilience (Pettit et al., 2010). In doing so, this study demonstrates how frontline actor adaptability drives systemic renewal in supply chains. Furthermore, the study methodologically contributes by illustrating the utility of partial least squares structural equation modeling (PLS-SEM) in analyzing multilevel TSCR constructs, following best practices for complex model testing (Hair et al., 2017; Mirzabeiki & Aitken, 2023). The COVID-19 pandemic, the acceleration of digital transformation, ongoing geopolitical tensions (e.g., China-U.S. trade disruptions), and increasing pressure for sustainable operations have collectively highlighted the fragility of global logistics networks (He et al., 2023; Mollenkopf et al., 2024). In this context, the role of HR-SCM practices becomes especially critical, not only for maintaining continuity but also for embedding adaptive and transformative capabilities into frontline operations. AI readiness further intensifies this shift, acting as both an enabler and a challenge for workforce and process integration. To clarify exactly where workforce microfoundations are embedded, we explicitly model HR-SCM practices as routinized mechanisms through which logistics firms build resilience: Cross-training expands resource flexibility, flexible shift design increases temporal flexibility, and managerial support strengthens coordination and resource access. These workforce-level mechanisms constitute the microfoundations that enable dynamic reconfiguration and disruption response in logistics operations (Felin et al., 2012; Teece, 2007b), consistent with resilience frameworks emphasizing adaptive capacity and recovery (Lengnick-Hall et al., 2011; Pettit et al., 2010).

LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

Foundations of TSCR

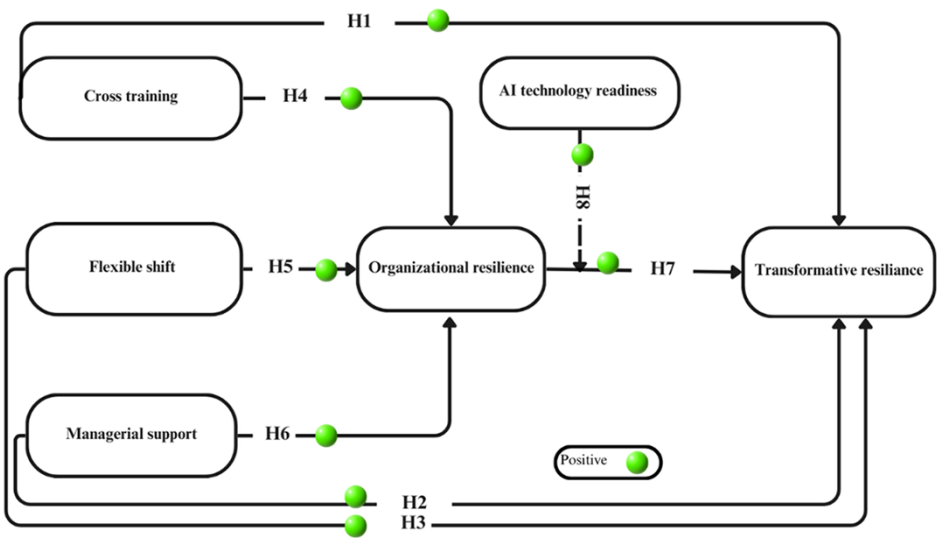
The shift to TSCR necessitates a popular approach to conducting and framing SCM research. To guide this shift, we build on existing transformative research and identify two key pillars of TSCR: the broadened SCM research ecosystem and the need for innovative research frameworks (Teece, 2007b). By integrating HR-SCM practices with AI technology readiness, this study advances a human-technology complementarity perspective of resilience. Building on the dynamic capabilities theory (Teece et al., 1997) and supply chain resilience frameworks, we conceptualize transformative

resilience as a human-centered adaptive process that enables employees to thrive in digitally augmented work systems (He et al., 2023; Teece, 2007b).

The supply chain is, in essence, an ecosystem rooted in systems thinking, viewing it as a dynamic network of exchanges and interactions among actors (Bille et al., 2025; Spohrer et al., 2007). Systems thinking is used in TSCR to study the social dimension of supply chains as related to individual and collective resilience. Researchers should also turn their attention not only to the traditional focus actors in supply chains but also to their upstream suppliers and downstream customers, the modern supply chain ecosystem (Bille et al., 2025). Today's new generation relies on a broader set of stakeholders, is more interdependent, and requires more direction (Mirzabeiki & Aitken, 2023). These parties face similar adaptation challenges and are expected to work together to develop novel solutions to address supply chain turbulence (Ketchen et al., 2014; Stolze et al., 2016).

In TSCR, the research should not be limited to concluding firm performance per se but, rather, to studying more generally how supply chains contribute to building resilience and transformativeness (Mirzabeiki & Aitken, 2023). To tackle these challenges, academics propose developing new models that account for AI readiness, workforce flexibility, and organizational adaptability. As digital technology and AI are increasingly used by businesses to improve their systems, these frameworks will enable the assessment of the supply chain's resilience, sustainability, and integration with AI (Wu et al., 2024). Integrating these aspects into supply chain studies, TSCR will help address issues beyond economic concerns and economic performance by analyzing how supply chains affect community support, sustainability culture-building, societal well-being, and adaptive capacity in a fast-changing business environment. The conceptual model is presented in Figure 1.

Figure 1. The Research Model



Note. AI = artificial intelligence.

Underpinning Theories

The study was developed based on TSCR (Pettit et al., 2010) and dynamic capabilities (Teece, 2007b) theories as a theoretical foundation for AI in organizations. From a resiliency perspective, advanced predictive analytics can enhance an organization's capacity to prevent, respond to, and recover from a disruption by continuously monitoring workforce performance signals, including

absenteeism trends and variations in throughput, and enabling early warning (Wu et al., 2024). From a dynamic capability perspective, AI tools, such as machine learning models and reinforcement learning algorithms, operate as adaptive routines that can detect changes in labor dynamics (Felin et al., 2012), suggest ways to optimize cross-training, and reorganize frontline practices on the fly.

HR-SCM Practices as Workforce Microfoundations of Resilience in Logistics

In logistics operations, resilience is not produced only by structural buffers (e.g., redundancy) but also by workforce-enabled adaptation (Felin et al., 2012), particularly in disruption-heavy environments where response quality depends on frontline decisions, rapid coordination, and flexible task execution (Khanuja & Jain, 2022). Therefore, we conceptualize HR-SCM practices as microfoundations that shape the routines through which firms sense disruptions, reallocate labor, and reconfigure operating processes. This logic is consistent with the microfoundations view that organizational capabilities emerge from individual- and team-level actions and routines (Felin et al., 2012). From a dynamic capabilities perspective, HR-SCM practices help operationalize sensing, seizing, and reconfiguring by building workforce versatility (cross-training), temporal flexibility (shift design), and resource access/coordination (managerial support) (Teece, 2007b). From a supply chain resilience perspective, these practices increase the organization's capacity to absorb shocks and recover by improving flexibility and responsiveness in the logistics node that connects multiple supply-chain partners (Pettit et al., 2010). This view also aligns with HRM (Human Resources Management) research showing that resilience capacity can be developed through strategic human resources systems that cultivate adaptability, learning, and coordinated action under stress (Lengnick-Hall et al., 2011). Accordingly, our model positions cross-training, flexible shift design, and managerial support as HR-SCM levers that enable organizational resilience and, through it, transformative resilience among logistics employees.

Cross-Training and Transformative Resilience

Cross-training is essential in fostering transformative resilience, as it enhances workforce adaptability and enables organizations to adjust rapidly to disruptions, aligning with the theories of dynamic capabilities and supply chain resilience. Dynamic capabilities were described by Teece (2018) as the firm's ability to sense and seize opportunities while reconfiguring its resource base to innovate or adapt. Cross-training enhances this capability, which enables people to meet a company's fluctuating workload demands by being prepared to support all roles in the organization. Likewise, with Supply Chain Resilience, as Zhang et al. (2023) noted, network resilience is the ability of an organization to prevent, withstand, respond to, and recover from disruptions, with the capability to continue operations and recover. By enhancing workforce flexibility and enabling rapid labor reallocation during crises, cross-training directly contributes to transformative resilience, allowing organizations not only to survive but to thrive amidst uncertainty (Mirzabeiki & Aitken, 2023). Therefore, cross-training serves as a strategic tool that operationalizes resilience by embedding adaptability into everyday practices, which is crucial for long-term sustainability and competitiveness in the face of evolving challenges. Hence, we proposed:

H1. Cross-training positively impacts transformative resilience.

Managerial Support and Transformative Resilience

Managerial support is a critical enabler of transformative resilience, as it provides the resources, guidance, and organizational stability necessary for employees to navigate disruptions and drive change (Soomro & Khan, 2024). From the dynamic capabilities perspective, as defined by Teece (2018), managerial support facilitates the sensing and seizing of opportunities by equipping employees with the resources and autonomy needed to adapt to evolving situations. When managers offer clear

guidance and foster a culture of open communication, they empower employees not only to respond to disruptions but also to innovate and transform processes in real time, which are key elements of transformative resilience. In the context of supply chain resilience, managerial support aligns with Pettit et al.'s (2010) concept of resilience, which is the capacity to absorb shocks, adapt, and recover. Mirzabeiki and Aitken (2023) found that supportive management practices provide employees with the emotional and operational support needed to overcome challenges, thereby enhancing their ability to recover and thrive under adverse conditions. Therefore, managerial support acts as a driver of transformative resilience by reinforcing employees' adaptive capacity, psychological flexibility, and resourcefulness, allowing them to navigate and learn from disruptive events effectively (Soomro & Khan, 2024). Hence, we hypothesized the following.

H2. Managerial support positively impacts transformative resilience.

Flexible Shift Design and Transformative Resilience

Flexible shift design is a key factor in fostering transformative resilience, as it enables employees to quickly adapt to changing operational requirements and external disruptions. According to the dynamic capabilities theory (Teece, 2018), flexibility in work schedules allows an organization to sense and seize opportunities in real time, which is crucial for transformative resilience. Flexible shifts allow workers to maintain productivity during disruptions, adapt to fluctuating demands, and continuously improve operational effectiveness. Additionally, from the perspective of supply chain resilience, as defined by Pettit et al. (2010), flexible scheduling enables better responsiveness and recovery during crises, ensuring the workforce can meet changing demands without compromising its well-being. Therefore, flexible shift design directly contributes to the workforce's ability to adapt and transform in response to challenges, enhancing transformative resilience. Hence, we proposed the following hypothesis.

H3. Flexible shift design positively influences transformative resilience.

Cross-Training and Organizational Resilience

Cross-training is essential for enhancing organizational resilience, as it enables the workforce to handle a variety of tasks and adapt to changing roles when needed (Nembhard, 2014). By referring to dynamic capabilities theory (Teece, 2007), cross-training strengthens an organization's ability to reconfigure resources in response to disruptions and uncertainties. By broadening employees' skill sets, organizations can more easily allocate resources where they are needed most during crises, thereby improving recovery and continuity (Lengnick-Hall et al., 2011). Pettit et al. (2010) highlight that organizational resilience is built on proactive measures that enable firms to anticipate and recover from disruptions. Cross-training plays a crucial role in preparing employees to assume unfamiliar roles, thereby improving the organization's ability to remain functional during crises and recover more effectively (Nembhard, 2014; Slomp & Molleman, 2002). Therefore, we proposed the following hypothesis.

H4. Cross-training positively impacts organizational resilience.

Flexible Shift Design and Organizational Resilience

Flexible shift design enhances organizational resilience by providing employees with the adaptability needed to respond to operational challenges (Karman, 2020). As noted in the dynamic

capabilities theory (Teece, 2007b), organizations must be able to sense and seize new opportunities to maintain performance under changing conditions. Flexible shifts support this capability by enabling the organization to adjust its workforce availability according to operational requirements (Perez-Valls et al., 2016). Pettit et al. (2010) define organizational resilience as the capacity to absorb and recover from disruptions. Flexible work schedules enable organizations to respond quickly and effectively to unforeseen events, ensuring the continuous flow of operations even during disruptions (Shifrin & Michel, 2022). Therefore, we proposed the following hypothesis.

H5. Flexible shift design positively impacts organizational resilience.

Managerial Support and Organizational Resilience

Managerial support is a crucial factor in fostering organizational resilience (Karman, 2020), as it provides employees with the guidance and resources necessary to navigate challenges effectively. From the perspective of dynamic capabilities (Teece, 2007b), supportive management practices enable organizations to quickly reconfigure and adapt by offering clear directions and ensuring access to resources during times of uncertainty (Shah et al., 2022). Managerial support contributes to resilience because it creates a reliable internal environment for adaptation (Lengnick-Hall et al., 2011; Wieland & Durach, 2021). Supportive leaders reduce uncertainty by clarifying priorities and removing operational barriers (Eisenberger et al., 1986; Teece, 2007b). This improves coordination and enables faster reallocation of labor and resources during disruptions (Ambulkar et al., 2015). In supply chains, such internal flexibility is repeatedly identified as a foundation for recovery and continuity (Pettit et al., 2010). Therefore, we proposed the following hypothesis.

H6. Managerial support positively impacts organizational resilience.

Organizational Resilience and Transformative Resilience

Organizational resilience plays a critical role in shaping transformative resilience by providing the foundation for employees to adapt, learn, and grow in response to disruptions (He et al., 2023). Resilience provides the organizational conditions that enable individuals to adapt and improve after disruptions. Supply chain resilience frameworks describe organizational resilience as the capacity to prepare for, respond to, and recover from shocks while maintaining performance (Brandon-Jones et al., 2014; Pettit et al., 2010). Dynamic capabilities theory further explains why this matters. Firms that can reconfigure routines and redeploy resources can turn disruption into an opportunity for renewal (Schilke et al., 2018; Teece, 2007b). These organizational actions shape how employees respond. Employees are more likely to learn, innovate, and remain resourceful when the organization supports adaptive routines (Duchek, 2020; Lengnick-Hall et al., 2011). As a result, organizational resilience should strengthen employees' transformative resilience. Hence, the following hypothesis is proposed.

H7. Organizational resilience positively impacts transformative resilience

The Moderating Role of AI Technology Readiness

AI technology readiness can be viewed as a boundary condition that enables organizations to operationalize resilience through technology-supported routines (Parasuraman, 2000). In resilient logistics organizations, employees need timely information, decision support, and coordination to respond effectively to disruptions (Pettit et al., 2010). Dynamic capabilities theory argues that firms achieve adaptation when they can reconfigure resources and routines as conditions change (Teece,

2007b; Teece et al., 1997). AI tools can accelerate this process by strengthening sensing and rapid response, but only when the organization is ready to integrate such tools into work practices (Wu et al., 2024). At the microlevel, readiness supports the translation of organizational routines into employee learning and innovation, which are central elements of transformative resilience (Felin et al., 2012; Lengnick-Hall et al., 2011). Accordingly, higher AI readiness should strengthen the relationship between organizational resilience and transformative resilience. Therefore, we proposed the following hypothesis.

H8. AI technology readiness moderates the relationship between organizational resilience and transformative resilience.

METHODOLOGY

Research Context

This study selected the logistics sector in China as its research context; the state of this sector is arguably among the most advanced in terms of technology adoption, serving as a linchpin of various global supply chains. As one of the largest logistics markets globally, China plays a crucial role in trade and supply chain operations (Li et al., 2022). There are numerous small firms in the logistics industry across China, located in various regions, from major cities to rural areas, providing an ideal environment for investigating workforce resilience.

In the logistics industry, workforce resilience should be a key focus, as it is crucial to the efficient and effective operation of the supply chain (Liu, W. et al., 2022). However, little research has examined this concept in relation to employee behavior. Given the dynamic nature of logistics operations, widely characterized in industry practice, employees are increasingly required to adopt new technologies or purposes, cope with rapidly increasing demands, and confront supply chain vulnerabilities (Yang & Yu, 2025). This research reflects the significance of addressing these struggles. Post-pandemic implications have also been raised in the logistics industry, e.g., workers' flexibility and the mental health of some logistics staff in a world that is gravitating toward the implementation of AI technology and remote working environments (He et al., 2023).

Truck drivers and sorting-hub workers in the Chinese logistics industry represent a unique and challenging case of organizational resilience and transformative resilience. Our research in this area also sought to address this need while offering a better understanding of how HR-SCM practices may positively affect a firm's transformative resilience, and reflecting on the broader implications of doing so in an innovative manner. Since these are essential elements of technology readiness and management support, this knowledge is critical.

Pretest and Procedures

This study examined the relationships between HR-SCM practices and employee resilience among truck drivers and sorting hub workers at Chinese logistics firms. A questionnaire was developed to collect data regarding organizational resilience, transformative resilience, and preparedness for AI technology. Nine content experts reviewed the questionnaire. Five experts had extensive experience in logistics operations and HR management within the Chinese logistics sector. In their turn, the four remaining experts were scholars with profound expertise in SCM and organizational behavior. The experts were asked to assess the clarity, ambiguity, and readability of the survey instrument's questions. After the detailed evaluation of the proposed questionnaire, the experts concluded that it is straightforward to understand. The unit of analysis in this study is the individual frontline logistics employee. Given their proximity to operational disruptions and direct engagement with HR-SCM

practices, employees' perceptions provide valid insights into organizational and transformative resilience processes (Ambulkar et al., 2015; Pettit et al., 2010).

Nevertheless, they made minor suggestions on two questions to ensure they are appropriately tailored to the Chinese workforce. In the questions regarding AI technology readiness and managerial support, revisions were suggested to provide more relevant and precise examples that align with logistics workers' experiences. For instance, the wording of the following questions was slightly revised: "I feel comfortable using AI technology at my work," and "my manager provides me with enough support to use AI technologies to be more productive."

Sampling and Data Collection

This study focused on logistics employees in Chinese logistics firms, specifically truck drivers and sorting-hub workers in Greater China (including major urban hubs, such as Beijing, Shanghai, and Guangzhou). Frontline logistics employees, including truck drivers and sorting hub workers, were selected as respondents because they are actively involved in real-time operational adaptation and HR-SCM implementation. This approach aligned with prior resilience research emphasizing the importance of microlevel behavioral data (Wieland & Durach, 2021).

This study employed a purposive sampling approach targeting frontline logistics personnel, including truck drivers and sorting hub workers, as they are central to implementing operational flexibility and adaptive responses during disruptions (Ambulkar et al., 2016). These respondents interact closely with HR-SCM practices, such as cross-training and shift adjustments, making them suitable for evaluating resilience-related constructs. Although some respondents held higher educational qualifications, this reflects the evolving workforce composition in China's modern logistics sector, where automation and AI integration demand higher skill levels. The final sample of 280 respondents was verified for adequacy using G*Power 3.1, confirming sufficient statistical power (Hair et al., 2017). The data were also adequate to test the hypotheses; as our final $N = 280$ respondents exceeded the required minimum for adequate power, we had secured sufficient cases to be confident that testing statistical relations would yield valid results.

We used two criteria for inclusion: All the people surveyed had to be Chinese nationals and graduate, full-time employees in logistics. Participants received an explanation of the study objectives and provided verbal consent before data collection. A pretest was carried out with 42 participants to assess the clarity and reliability of the questionnaire. Once it was established to be valid, data collection was initiated.

Since this is a cross-sectional study, data were collected at a single point in time, providing a snapshot of employee behavior, organizational resilience, and AI readiness across the logistics industry. The survey was distributed to 329 employees (averaging 15 responses per firm) between May and July 2025. A total of 280 completed responses were received, yielding an 85.12% response rate after excluding nine invalid responses. Of these, 180 responses were collected through online surveys. To ensure accessibility and participation, we used Wenjuanxing, a widely accepted Chinese survey platform, alongside hard-copy surveys. The remaining 100 responses were collected through printed questionnaires distributed directly at logistics firms. Demographic information for the participants is presented in Table 1.

Table 1. The Respondent Profile

Category	Frequency	%
Gender		
Male	210	75.00%

continued on following page

Table 1. Continued

Category	Frequency	%
Female	70	25.00%
Age group (years)		
18–30	85	30.40%
31–40	110	39.30%
41–50	50	17.90%
51 and above	35	12.50%
Education level		
High school	30	10.70%
Bachelor's degree	120	42.90%
Master's/PhD	130	46.40%
Job role		
Truck driver	120	42.90%
Sorting hub worker	160	57.10%
Years of experience		
<1 year	45	16.10%
1–3 years	110	39.30%
3–5 years	75	26.80%
>5 years	50	17.90%
AI technology readiness		
Low	50	17.90%
Medium	140	50.00%
High	90	32.10%

Note. Source(s): Authors' own creation. AI = artificial intelligence.

Instruments

To measure the key variables in this study, well-established scales were adapted from the literature to fit the context of the Chinese logistics industry. The scales were refined to assess the relationship between HR-SCM practices, organizational resilience, and transformative resilience. A pilot test was conducted, and the scales' reliability was found to be satisfactory. Employee cross-training was measured using four items adapted from Ambulkar et al. (2016). One example item is: "I am trained to perform multiple roles within my organization, beyond my core function." This assesses the level of workforce flexibility achieved through cross-training initiatives. Flexibility in work schedules was assessed using four items, including: "I have the flexibility to adjust my working hours based on operational requirements." This item measures employees' control over their schedules, a key component of workforce resilience in the logistics industry.

Measurements of managerial support were based on a 4-item scale, e.g., "My supervisor provides clear guidance and support when faced with operational challenges." This item reflects the extent to which employees perceive management support during disruptions, thereby contributing to the overall organizational resilience. The measurement of organizational resilience was based on the following item: "Our organization quickly adapts to unexpected changes in the external environment (e.g., market shifts, crises)." AI readiness was evaluated using a 4-item scale, e.g., "I am

comfortable using AI tools and technologies that are introduced in my place of work." This item indicates employees' readiness to adopt AI technologies to enhance operational efficiency and make organizations more resilient. As for the measurements of transformative resilience, it was considered through dimensions, such as adaptive capacity, psychological flexibility, learning and innovation, resourcefulness, and growth following disruption. An example of an item measuring adaptive capacity is: "I can adjust my strategies when faced with unexpected challenges quickly," which demonstrates the extent to which employees can modify their strategies or plans in response to logistics glut, a factor crucial for workforce resilience. In this study, these scales and items were adapted or refined to ensure relevance and clarity in referring to logistics employees in Chinese logistics organizations. Table 2 presents the final set of items and scales.

Table 2. Convergent Reliability Analysis

Construct	FL	α	CR	AVE
CT1. I am trained to perform multiple roles within my organization, beyond my core function.	0.824	0.8899	0.932	0.71
CT2. I regularly assist colleagues in other departments when needed.	0.828			
CT3. Cross-training is actively encouraged within my organization to improve workforce flexibility.	0.832			
CT4. I feel confident in performing tasks outside of my usual responsibilities when required.	0.816			
FS1. I have the flexibility to adjust my working hours based on operational requirements.	0.821	0.921	0.912	0.73
FS2. The organization allows workers to change their shifts in the event of unexpected situations.	0.817			
FS3. I have control over my work schedule when there are sudden disruptions in operations.	0.823			
FS4. Flexible work hours are a standard practice within my organization, especially during busy periods.	0.824			
MS1. My supervisor provides clear guidance and support when faced with operational challenges.	0.825	0.901	0.942	0.76
MS2. Management encourages open communication about work-related difficulties and solutions.	0.828			
MS3. My supervisor ensures I have the resources necessary to perform my duties during disruptions.	0.829			
MS4. I feel that the leadership in my organization genuinely cares about employees' well-being.	0.827			
OR1. Our organization quickly adapts to unexpected changes in the external environment.	0.822	0.913	0.913	0.74
OR2. The organization is well-prepared to handle disruptions, and recovery plans are in place.	0.824			
OR3. The organization continuously evaluates its processes to identify opportunities for improving its handling of future disruptions.	0.825			
OR4. Our company learns from past challenges and applies the lessons to improve operations.	0.827			
TR1. I am comfortable using AI tools and technologies introduced at my workplace.	0.832	0.912	0.914	0.81
TR2. I believe that AI technologies can improve my job performance and efficiency.	0.829			

continued on following page

Table 2. Continued

Construct	FL	α	CR	AVE
TR3. I am confident in using AI-based solutions for managing tasks that were previously manual.	0.833			
TR4. The organization supports the integration of AI technology to enhance operational effectiveness.	0.831			
Transformative resilience				
AC1. I can quickly adjust my strategies when faced with unexpected challenges.	0.825	0.915	0.913	0.85
AC2. When confronted with setbacks, I can quickly shift my approach to find solutions.	0.828			
PF1. I can maintain my focus on my goals even when faced with adversity.	0.832	0.931	0.912	0.71
PF2. I stay calm and composed when unexpected disruptions occur.	0.827			
LI1. I actively seek new ways to improve my performance after a disruption.	0.831	0.912	0.901	0.82
LI2. I view challenges as opportunities to learn and become more effective.	0.826			
RF1. I am able to find alternative solutions when the original plan doesn't work.	0.829	0.889	0.921	0.74
RF2. During crises, I use all available resources creatively to solve problems.	0.832			
GPD1. I feel more capable and confident after overcoming a challenging situation.	0.824	0.912	0.901	0.85
GPD2. I believe that overcoming difficulties has made me stronger and more resilient.	0.823			

Note. Source(s): Authors' creation. FL = Factor Loading; CR = composite reliability; AVE = average variance extracted; AI = artificial intelligence; CT = cross-training; FS = flexible shift design; MS = managerial support; OR = organizational resilience; TR = AI technology readiness; AC = adaptive capacity; PF = psychological flexibility; LI = learning and innovation; RF = resourcefulness; GPD = growth post-disruption.

Analytical Strategy

Exploratory factor analysis was used as an initial step to examine the factor structure of each multi-item scale and determine whether the items loaded on their allotted constructs. Subsequently, confirmatory factor analysis was employed to evaluate measurement validity (convergent and discriminant) and to test the fit of the proposed measurement model. After validating the measurement model, the structural relationships among the constructs were tested using PLS-SEM. It is crucial that PLS-SEM could be used to test direct effects, the mediating role of organizational resilience, and the moderating effects of tenure and AI technology readiness on the relationship between organizational resilience and transformative resilience. All analyses were consistent with Hair et al.'s (2019) guidelines to ensure the quality of the model and significance of the paths.

RESULTS

Analytical Techniques

Our study employed PLS-SEM using SmartPLS (Version 4) to analyze the measurement and structural models. PLS-SEM was used over Covariance-Based structural equation modeling for several reasons. First, PLS-SEM is nonparametric, meaning the data do not need to be normally distributed. This was particularly important for this study, as the data on the logistics workforce may not follow normal distribution patterns. Second, the method is predictive, which is essential for assessing the impact of HR-SCM practices on transformative resilience and other dependent variables (Hair et al., 2021). This predictive orientation aligns with the study's goal of identifying how HR-SCM practices and AI readiness contribute to transformative resilience. Unlike CB-SEM, which is confirmatory and theory-testing, PLS-SEM emphasizes theory development and prediction (Shmueli et al., 2019). Third,

PLS-SEM is particularly useful for relatively small sample sizes. The data were collected from a small sample of the company's employees. Lastly, if the analysis were predictive, PLS-SEM would be a superior method to CB-SEM, as it provides the most accurate estimates of path relationships. A consistent PLS approach was applied to estimate the measurement models, and bootstrapping with 5,000 resamples was used to assess the significance of the path coefficients and to calculate confidence intervals. To test the moderating effect of the AI technology readiness, the moderation slope plots were used. Moreover, an importance-performance map analysis was conducted to investigate the relative importance of each construct in predicting workforce resilience. All measurement–respondent alignments and model fit indices (R^2, f^2, Q^2, standardized root mean square residual) are reported to ensure transparency and methodological rigor.

Measurement Estimation

To assess potential common method bias, this study used the marker-variable technique recommended by Lindell and Whitney (2001), which is considered more robust than the Harman single-factor test. A theoretically unrelated construct—employees' preferred communication channel (e.g., preference for digital versus face-to-face communication)—was included as the marker variable, as it has no conceptual relationship with the core study constructs. The partial correlation adjustment procedure was applied by comparing the average correlations among substantive constructs in the original model with those in the marker-adjusted model. The difference between the two (Δr = 0.028) was statistically negligible, confirming that common method bias did not substantially affect the parameter estimates.

Furthermore, to ensure the dimensional integrity of the measures, factor loadings, composite reliability, Cronbach's alpha (α), and average variance extracted were examined following Hair et al. (2019). As shown in Table 2, all values exceeded the recommended thresholds (composite reliability > 0.70, average variance extracted > 0.50), indicating strong convergent validity for the measurement model.

Additionally, we assessed discriminant validity by calculating the heterotrait-monotrait ratio. As shown in Table 3, discriminant validity is supported, with all heterotrait-monotrait values below the recommended threshold of 0.85 (Becker et al., 2023). This confirms that the constructs in our model are distinct and adequately measured. Model adequacy was assessed through R^2, Q^2, and SRMR indices, all within recommended thresholds.

Table 3. The Descriptive Statistics and Discriminant Validity Analysis

Constructs	Discriminant Validity					Descriptive Statistics			
	1	2	3	4	5	Mean	STDDEV	Skewness	Kurtosis
1. Cross-training	0.491	0.410	0.373	0.483	0.433	3.89	1.213	-0.782	-0.567
2. Flexible shift	0.423	0.463	0.452	0.482	0.445	4.12	1.134	-0.665	-0.452
3. Managerial support	0.496	0.565	0.462	0.512	0.481	4.23	1.198	-0.609	-0.329
4. Organizational resilience	0.472	0.485	0.492	0.526	0.475	4.56	1.231	-0.563	-0.375
5. Transformative resilience	0.433	0.466	0.473	0.481	0.565	4.40	1.154	-0.688	-0.412

Note. Source(s): Authors' own creation. STDDEV = standard deviation.

Hypothesis Testing

The PLS-SEM results confirmed that all proposed hypotheses in our model were supported (see Table 4). H1 demonstrated that cross-training positively affects organizational resilience ($\beta = 0.415$, $t = 6.837$), highlighting that cross-training enhances employees' adaptability and effectiveness during

disruptions. H2 found a significant positive relationship between managerial support and organizational resilience (β = 0.512, t = 7.562), indicating that supportive management practices are essential for fostering resilience. H3 showed that managerial support also positively influences transformative resilience (β = 0.368, t = 5.284), suggesting that it contributes to employees' growth and recovery during challenging times. In H4, cross-training was found to significantly enhance transformative resilience (β = 0.398, t = 6.342), confirming that it not only boosts organizational resilience but also contributes to personal growth. H5 supported the notion that flexible shift design has a positive impact on organizational resilience (β = 0.356, t = 5.870), emphasizing the role of flexible work arrangements in enabling employees to adapt to disruptions. H6 revealed that managerial support positively affects transformative resilience (β = 0.321, t = 5.129), reinforcing the idea that strong leadership fosters resilience.

Table 4. The Results of Hypothesis Testing

Relationship	**β**	***T*-Value**	***P*-Value**	**Effect Size (f^2)**	**Coefficient of Determination (R^2)**	**Geisser (Q^2)**	**Remark**
Direct effects							
H1: Cross-training → organizational resilience	0.415	7.234	0.000	0.258	0.236	0.174	Supported
H2: Managerial support → organizational resilience	0.512	8.901	0.000	0.324	0.247	0.169	Supported
H3: Managerial support → transformative resilience	0.368	6.41	0.000	0.274	0.212	0.151	Supported
H4: Cross-training → transformative resilience	0.398	6.82	0.000	0.289	0.246	0.158	Supported
H5: Flexible shift design → organizational resilience	0.356	6.22	0.000	0.231	0.239	0.163	Supported
H6: Managerial support → transformative resilience	0.321	5.39	0.000	0.214	0.219	0.138	Supported
H7: AI technology readiness → moderating effect on organizational resilience → transformative resilience	0.287	5.712	0.000	0.203	0.181	0.132	Supported
moderation							
Relationship	β	STDDEV	*T*-Value	*P*-Value	Remark		
Cross-training → transformative resilience	0.225	0.038	5.93	0.000	Supported		
Flexible shift → transformative resilience	0.261	0.042	6.211	0.000	Supported		
Managerial support → transformative resilience	0.349	0.037	9.493	0.000	Supported		
Organizational resilience → transformative resilience	0.329	0.032	10.25	0.000	Supported		
AI technology readiness × cross-training → transformative resilience (H7a)	0.138	0.034	4.051	0.000	Supported		

continued on following page

Table 4. Continued

Relationship	β	*T*-Value	*P*-Value	Effect Size (f^2)	Coefficient of Determination (R^2)	Geisser (Q^2)	Remark
AI technology readiness × flexible shift → transformative resilience (H7b)	0.157	0.039	4.115	0.000	Supported		
AI technology readiness × managerial support → transformative resilience (H7c)	0.198	0.043	4.611	0.000	Supported		

Note. Source(s): Authors' own creation. AI = artificial intelligence; STDDEV = standard deviation.

All indirect and moderating effects were tested using 5,000 bootstrap resamples, ensuring robustness of the mediation (organizational resilience) and moderation (AI technology readiness) paths. Interaction effects were further visualized using simple slope plots, which confirmed the significance and direction of the moderation effects (Hair et al., 2017). H7 found that AI technology readiness moderates the relationship between organizational and transformative resilience ($\beta = 0.287$, $t = 4.762$), underscoring AI readiness's significant role in amplifying the positive effects of organizational resilience. These findings provide strong support for the theoretical framework, demonstrating that HR-SCM practices and AI readiness are critical drivers of organizational and transformative resilience in logistics firms.

Fit Index

Cohen's f^2 values ranged between small and medium effects (see Table 4). The coefficient of determination (R^2) indicates that the model explains 24.5%, 29.8%, and 62.3% of the variation in organizational resilience, transformative resilience, and AI technology readiness, respectively. Additionally, Stone-Geisser (Q^2) values for the endogenous variables, organizational resilience, transformative resilience, and AI technology readiness, were 0.175, 0.165, and 0.342, respectively, indicating that the model has predictive relevance (Hair et al., 2019).

Moderation Effect Testing

The results indicate that AI technology readiness significantly moderates the relationship between HR-SCM practices (cross-training, flexible shift design, and managerial support) and transformative resilience. Specifically, cross-training had a positive effect on transformative resilience ($\beta = 0.225$, $t = 6.325$, $p < .001$), and AI readiness strengthened this path. The positive relationship of Flexible Shift Design and *m*-resilience was enhanced by AI technology readiness ($\beta = 0.256$, $t = 5.921$, $p < 0.001$). Furthermore, managerial support also had a significant effect on adaptive resilience ($\beta = 0.349$, $t = 9.558$, $p < 0.001$), and this relationship was again moderated by readiness for AI. These results imply that AI readiness plays a vital role in supporting the effectiveness of HR-SCM practices in achieving transformational resilience. The moderating effects of these relationships were significant, confirming the hypotheses that AI readiness enhances the influence of HR-SCM practices on transformative resilience.

DISCUSSION

The goal of this paper was to empirically test the impact of HR-SCM practices on organizational and transformative resilience among logistics service providers. Building on the dynamic capabilities and supply chain resilience literature, we empirically tested the roles of cross-training, managerial support, and flexible scheduling shifts in strengthening resilience, leveraging AI readiness as an

enabler. These results are consistent with previous studies that emphasize the role of workforce flexibility and resource deployment in responding to disruptions (Pettit et al., 2010; Teece, 2007b).

To be quick and flexible in addressing potential problems, every organization implements staff cross-training. This allows the team to respond to nonstandard problems (Zhang et al., 2023). Moreover, a manager's role is also a necessary mechanism for resisting disturbances through the institution of leadership forms (Teece, 2018). It confirms a point made in Teece (2018): Central managers' guidance is necessary for enterprises to have both supply chain resilience and adaptive capacity. This paper also illustrates the benefits of flexible shifts and offers businesses a method to reduce disruptions while allowing workers to control their day-to-day lives. Consistent with dynamic capabilities theory, organizations increase resilience by purposefully assembling and deploying resources over time and digital preparedness, operationalized as AI readiness, acts as a moderating variable in this process. In addition to promoting resilience outcomes, this means we need digital preparedness, and AI technology readiness serves as a moderating variable in the process just described. We can conclude that organizations can use AI as a strategic partner for organizational resilience and transformation, as well as for day-to-day decision-making through central forecasting (He et al., 2023; Mollenkopf et al., 2024). The results confirm that HR-SCM practices act as dynamic enablers that strengthen organizational resilience, thereby fostering transformative resilience among logistics employees. This causal pathway aligns with the dynamic capabilities perspective (Teece, 2007b), wherein adaptive human resource systems serve as microfoundations for organizational learning and renewal (Ambulkar et al., 2015). Consistency across hypotheses, statistical estimates, and theory supports the robustness of the proposed model.

From a practical perspective, our findings have important implications for logistics managers seeking to improve organizational workforce flexibility. When reimagining HR-SCM strategies to establish an agile labor force that can duplicate activities when disrupted, it is essential to focus on cross-training and management support. Flexible working and an adaptable shift pattern are effective ways to ensure your logistics operations continue to run. Matching business with the level of AI is also essential for companies that aim to increase resilience through digital technology, as AI has strong potential to enhance sensing capabilities, secure power, and transform how an organization responds to disturbances (Liang et al., 2025).

Our findings offer timely implications for logistics firms navigating AI adoption, shifting labor conditions, and volatile market environments. In an era where disruptions are the norm, rather than the exception, developing human-centric resilience, through HR-SCM practices, complements digital investments and strengthens long-term operational agility (Wieland & Durach, 2021). By studying these factors from a logistics perspective in China, our research provides significant insights into the overall area of TSCR. The combination of workforce agility, HR practices, and AI readiness offers a comprehensive framework for establishing resilient supply chains. Amid the ongoing chaos around us, this research demonstrates that companies need to reevaluate their supply chains beyond operational efficiency, considering resilience and flexibility as the real modus operandi for future success.

Theoretical Implications

This study contributes to the theoretical aspects of HR-SCM practices (such as dynamic capability, strategic alignment, and real-time knowledge sharing) in conjunction with organizational resilience and transformative resilience in logistics firms. The model has several implications for workforce theory and practice. First, by investigating how cross-training, managerial support, and flexible shift design build adaptive capacity in the workforce, it helps firms sense disruptions quickly, seize opportunities effectively, and transform smoothly (Teece, 2007b), thereby extending dynamic capabilities theory. The study's findings underscore the necessity for these HR practices to enable organizations to seamlessly incorporate workforce flexibility into day-to-day operations (Pettit et al., 2010). Our findings contribute to the literature on the importance of workforce adaptability for

operational continuity, particularly in logistics firms, and provide further insight into how HR-SCM practices can impact organizational resilience.

Second, it extends the literature on transformative resilience by demonstrating that organizational resilience serves as a precursor to it. This essay utilizes Teece's (2018) dynamic capabilities framework as a lens to examine how resilient organizations not only bounce back but also transform and innovate in the face of emerging new challenges. We concluded that, when an organization is resilient, it helps create transformative resilience in the process, as its employees learn not only to adapt and recover from disruptions but also to use them as opportunities for change.

Furthermore, the study contributes to literature by investigating the moderating role of AI technology readiness in the relationship between organizational resilience and transformative resilience. The contributions of this paper are more general than those of Zhang et al. (2023). As an alternative source points out, further digitalization is critical to supply chains' capacity to adapt and refashion their operations using new technologies (The Economist Intelligence Unit, 2023). The study adds to the supply chain risk management literature by proposing an integrated model that embeds the relationship between AI readiness, organizational resilience, and transformative resilience within supply chain collaboration (SCC), in a way that has not been done before. In particular, the model foregrounds the digital dimension and incorporates emerging trends in the digitization of supply chains.

This article contributes to the literature by showing that workforce adaptability leads to organizational and transformative resilience, with HR-SCM practices mediating the relationship and AI readiness moderating it. Our findings provide empirical insights that are important for conducting a qualitative study on how logistics firms can use human resource management practices and digital techniques to enable organizations in ways that support sustainable, resilient operations. There are frequent disruptions to the normal functioning of global logistics and supply chains, upending traditional indicators of resilience, which must therefore be redefined or invented anew. In contrast, the thrust of our research is capacity building, not so much a lean, fit business as an enterprise's ability to adapt, flexibly change, and/or (re)invent things whenever the future appears uncertain.

Managerial Implications

The results of the present study show essential management implications for logistics firms attempting to increase their organizational resilience and that of their employees in general. These principles will enable managers to understand how HR-SCM practices (cross-training, managerial support, and flexible shift design) foster more flexible employees. In brief, stay ahead of the curve in reform and practice. It results in an organization that recovers quickly from shocks because its people are effectively placed in the right roles. Duties are about managing for an organizational feel, enabling flexible work, and (especially if the individual has any language skills relevant to other countries) multiskilling in an emergency. The majority of new businesses will require the office to exist and function in ways that are congenial to performing their tasks, cross-training across several outputs helps ensure that all companies, except specialized ones, will emerge more or less intact when shocks strike.

The second key adaptive role is AI tech preparedness for business continuity and resilience. As digital capabilities and AI become increasingly embedded in firms' operations, managers will need to train their workforce in AI, providing hands-on guidance so that it becomes an organizational culture for technology. Such green AI logistics firms will be able to respond even more nimbly and quickly to market conditions, providing real-time innovations and a flexible workforce that adapts promptly to ongoing innovations. Third, in connection with the observatory, both institutional insulation and transformative resilience require managerial backing. Especially these days, when everything feels a little blurry, managers should ask employees what they need to feel supported. The research suggests that greater trust and collaboration require open communication between management and staff, including regular feedback. The more we can make our employees feel secure in their roles, the more involved they will be, thereby achieving higher job satisfaction and organizational resilience over time.

Fourth, flexible shift design is a crucial HR intervention within HR-SCM that affects both organizational and transformative resilience. Logistics firms should introduce flexible shift options, allowing employees to choose their preferred timing slot based on lead requirements. Aside from making work-life balance easier for employees, this strategy enables the organization to adjust to shifting demands and unanticipated disruptions more readily. This part helps the business diversify much faster, making it more agile (very important for the survive/recover phase, as long as there is enough market demand) and actually feel safer across the industry they are working in. Aside from this, offering learning and development opportunities, mentor or coach support, and transparent career paths can help promote an employee's perceived employability so that they may stay with you for a long time in the future. Highly engaged, employable, and productive employees are resilient in the face of any challenge.

As logistics operations increasingly adopt AI and automation, fostering employee adaptability through HR-SCM practices becomes a strategic imperative. AI readiness amplifies this adaptability by enhancing human–machine collaboration, allowing organizations to respond more intelligently and flexibly to disruptions.

Limitations and Future Studies

While this study has important implications, it is limited by several factors. This study has the following limitations: First, it was a cross-sectional survey at a single point in time. Although this is a valuable technique for understanding variation, it is not helpful for inferring causation or the timing between variables. Although our cross-sectional design limits causal inference, theoretical consistency and statistical robustness strengthen confidence in the proposed relationships; future longitudinal studies could provide deeper insights into temporal causality between HR-SCM practices, organizational resilience, and transformative resilience.

Second, we consider HR-SCM practices (e.g., cross-training, managerial support, and flexible shift) as microfoundations of resilience. These practices are essential, but they exist within a larger formal organizational context. One area for further research is how HR practices and managerial components, such as leadership styles and organizational culture, can provide a comprehensive picture of how organizations build resilience in workplaces of uncertainty. Third, this study finds that AI technology readiness moderates the relationship between organizational resilience and transformative resilience. While the results are interesting, AI readiness is merely one pillar of digital transformation. The dynamics of other digital transformations, such as big data analytics and automation reconciled with HR-SCM practices, could be examined to reinforce resilience and innovation in supply chains, offering an avenue for future research.

Finally, our study focuses specifically on logistics firms in China, which provides rich insights but may also constrain the generalizability of the findings to other geographies. Future research may examine the effects of HR-SCM practices, organizational resilience, and AI technology readiness on resilience in different cultural or industry contexts (e.g., Europe and North America) to yield findings applicable across contexts. Finally, future research can further investigate how resilient culture and AI readiness influence organizational outcomes in different types of organizations (e.g., startups versus established ones) to understand the processes by which such factors contribute positively when embedded in various institutional contexts.

CONCLUSION

This study investigated HR-SCM practices and their role as enablers of organizational and transformational resilience in logistics companies, by focusing on the moderating effect of AI technology readiness. Cross-training, leadership support, and flexible shift patterns accounted for much of the variance in both organizational resilience and transformative resilience. Our results align with the theoretical underpinnings of DCs and resiliency, namely employee flexibility and

technological preparedness for disruptions. AI technology readiness moderates the relationship between organizational and transformative resilience, indicating a new stream of literature on the centrality of digital transformation to supply chains.

The current research makes several contributions to the transformative supply chain. First, the model proposed in this study is the first to connect workforce adaptability with HR practices and digital readiness and resilience outcomes. These new ways to shield from and recover quickly after a disruption build on AI technology readiness and align with HR-SCM practices. An even more significant contribution of our work is to contest the hitherto predominant emphasis on operational efficiency and propose resilience and transformation as key themes for long-term success. Second, our study provides several practical recommendations for logistics managers to achieve workforce resilience. For example, cross-training, managerial support, and work shift flexibility are three significant investments organizations can make, allowing their workers to protect themselves from disruptions while also creating new opportunities.

FUNDING STATEMENT

This research was supported by the Chinese Government Scholarship. The authors also acknowledge the support provided by Chang'an University in providing the facilities and resources necessary to conduct this study.

COMPETING INTERESTS

The authors of this publication declare that there are no competing interests.

CORRESPONDING AUTHOR

The corresponding author for this research is Aicha Ndiaye, who can be reached at aichawalyn@gmail.com or at the following postal address: Chang'an University, middle-section of Nan'er Huan Road Xi'an, ShaanXi Province, 710064, China.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their insightful comments which helped improve this manuscript. This research was conducted under the supervision of Dr. Li Yibin of Chang'an University.

REFERENCES

Ambulkar, S., Blackhurst, J. V., & Cantor, D. E. (2016). Supply chain risk mitigation competency: An individual-level knowledge-based perspective. *International Journal of Production Research*, *54*(5), 1398–1411. DOI: 10.1080/00207543.2015.1070972

Becker, J., Cheah, J., Gholamzade, R., Ringle, C. M., & Sarstedt, M. (2023). PLS-SEM's most wanted guidance. *International Journal of Contemporary Hospitality Management*, *35*(1), 321–346. DOI: 10.1108/IJCHM-04-2022-0474

Bille, A., Bjørndal Thomsen, K., Beske-Janssen, P., & Wieland, A. (2025). Understanding the political nature of supply chain management: A case study of the coffee supply chain. *Supply Chain Management*, *30*(2), 195–212. DOI: 10.1108/SCM-02-2024-0076

Brandon-Jones, E., Squire, B., Autry, C., & Petersen, K. (2014). A Contingent Resource-Based Perspective of Supply Chain Resilience and Robustness. *The Journal of Supply Chain Management*, *50*(3), 55–73. Advance online publication. DOI: 10.1111/jscm.12050

Dawood, H. M., & Afzal, N. (2025). Ethical issues and role of smart supply chain management. In Mubarik, M. S., & Khan, S. A. (Eds.), *Smart supply chain management: Design, methods and impacts* (pp. 73–97). Springer. DOI: 10.1007/978-981-96-1333-5_5

Duchek, S. (2020). Organizational resilience: A capability-based conceptualization. *Business Research*, *13*(1), 215–246. DOI: 10.1007/s40685-019-0085-7

Eisenberger, R., Huntington, R., Hutchison, S., & Sowa, D. (1986). Perceived organizational support. *The Journal of Applied Psychology*, *71*(3), 500–507. DOI: 10.1037/0021-9010.71.3.500

Esper, T. L., Clifford Defee, C., & Mentzer, J. T. (2010). A framework of supply chain orientation. *International Journal of Logistics Management*, *21*(2), 161–179. DOI: 10.1108/09574091011071906

Felin, T., Foss, N. J., Heimeriks, K. H., & Madsen, T. L. (2012). Microfoundations of routines and capabilities: Individuals, processes, and structure. *Journal of Management Studies*, *49*(8), 1351–1374. DOI: 10.1111/j.1467-6486.2012.01052.x

Hair, J. F.Jr, Hult, G. T. M., Ringle, C. M., Sarstedt, M., Danks, N. P., & Ray, S. (2021). *Partial least squares structural equation modeling (PLS-SEM) using R: A workbook*. Springer. DOI: 10.1007/978-3-030-80519-7

Hair, J. F., Hult, G. T. M., Ringle, C. M., Sarstedt, M., Richter, N. F., & Hauff, S. (2017). *A primer on partial least squares structural equation modeling*. Vahlen. DOI: 10.15358/9783800653614

Hair, J. F., Risher, J. J., Sarstedt, M., & Ringle, C. M. (2019). When to use and how to report the results of PLS-SEM. *European Business Review*, *31*(1), 2–24. DOI: 10.1108/EBR-11-2018-0203

Karman, A. (2020). Flexibility, coping capacity and resilience of organizations: Between synergy and support. *Journal of Organizational Change Management*, *33*(5), 883–907. DOI: 10.1108/JOCM-10-2019-0305

Ketchen, D. J.Jr, Wowak, K. D., & Craighead, C. W. (2014). Resource gaps and resource orchestration shortfalls in supply chain management: The case of product recalls. *The Journal of Supply Chain Management*, *50*(3), 6–15. DOI: 10.1111/jscm.12047

Khanuja, A., & Jain, R. K. (2022). The mediating effect of supply chain flexibility on the relationship between supply chain integration and supply chain performance. *Journal of Enterprise Information Management*, *35*(6), 1548–1569. DOI: 10.1108/JEIM-11-2020-0449

Lengnick-Hall, C. A., Beck, T. E., & Lengnick-Hall, M. L. (2011). Developing a capacity for organizational resilience through strategic human resource management. *Human Resource Management Review*, *21*(3), 243–255. DOI: 10.1016/j.hrmr.2010.07.001

Li, Q., Yan, R., Zhang, L., & Yan, B. (2022). Empirical study on improving international dry port competitiveness based on logistics supply chain integration: Evidence from China. *International Journal of Logistics Management*, *33*(3), 1040–1068. DOI: 10.1108/IJLM-06-2020-0256

Liang, Y., Liu, W., & Gong, Y. (2025). How does enterprise digital transformation impact supply chain resilience: A multi-case study from China. *Industrial Management & Data Systems*, *125*(11), 2909–2923. DOI: 10.1108/IMDS-12-2024-1243

Lindell, M. K., & Whitney, D. J. (2001). Accounting for common method variance in cross-sectional research designs. *The Journal of Applied Psychology*, *86*(1), 114–121. DOI: 10.1037/0021-9010.86.1.114 PMID: 11302223

Liu, W., Liang, Y., Bao, X., Qin, J., & Lim, M. K. (2022). China's logistics development trends in the post COVID-19 era. *International Journal of Logistics*, *25*(6), 965–976. DOI: 10.1080/13675567.2020.1837760

Mirzabeiki, V., & Aitken, J. (2023). Panarchy-based transformative supply chain resilience: The role of supply chain capital. *International Journal of Operations & Production Management*, *43*(1), 99–139. DOI: 10.1108/IJOPM-03-2022-0191

Mollenkopf, D. A., Esper, T. L., Stolze, H. J., & Ozanne, L. K. (2024). Transformative supply chain research: A new frontier for SCM scholars. *Journal of Business Logistics*, *45*(3), e12389. Advance online publication. DOI: 10.1111/jbl.12389

Nembhard, D. A. (2014). Cross training efficiency and flexibility with process change. *International Journal of Operations & Production Management*, *34*(11), 1417–1439. DOI: 10.1108/IJOPM-06-2012-0197

Parasuraman, A. (2000). Technology readiness index (TRI): A multiple-item scale to measure readiness to embrace new technologies. *Journal of Service Research*, *2*(4), 307–320. DOI: 10.1177/109467050024001

Pederneiras, Y. M., Meckenstock, J., Carvalho, A. I. C., & Barbosa-Póvoa, A. P. (2022). The wicked problem of sustainable development in supply chains. *Business Strategy and the Environment*, *31*(1), 46–58. DOI: 10.1002/bse.2873

Perez-Valls, M., Cespedes-Lorente, J., & Moreno-Garcia, J. (2016). Green practices and organizational design as sources of strategic flexibility and performance. *Business Strategy and the Environment*, *25*(8), 529–544. DOI: 10.1002/bse.1881

Pettit, T. J., Fiksel, J., & Croxton, K. L. (2010). Ensuring supply chain resilience: Development of a conceptual framework. *Journal of Business Logistics*, *31*(1), 1–21. DOI: 10.1002/j.2158-1592.2010.tb00125.x

Schilke, O., Hu, S., & Helfat, C. (2017). Quo Vadis, Dynamic Capabilities? A Content-Analytic Review of the Current State of Knowledge and Recommendations for Future Research. *The Academy of Management Annals*. Advance online publication. DOI: 10.5465/annals.2016.0014

Shah, S. A., Asif, M. A., Shoukat, M. H., Polatci, S., & Rehman, S. U. (2022). Quality management practices and inter-organizational project performance: Moderating effects of inter-organizational communication, relationship, and process conflicts in healthcare. *SAGE Open*, *12*(3). Advance online publication. DOI: 10.1177/21582440221113829

Shifrin, N. V., & Michel, J. S. (2022). Flexible work arrangements and employee health: A meta-analytic review. *Work and Stress*, *36*(1), 60–85. DOI: 10.1080/02678373.2021.1936287

Shmueli, G., Ray, S., Estrada, J. M. V., & Chatla, S. B. (2016). The elephant in the room: Predictive performance of PLS models. *Journal of Business Research*, *69*(10), 4552–4564. DOI: 10.1016/j.jbusres.2016.03.049

Slomp, J., & Molleman, E. (2002). Cross-training policies and team performance. *International Journal of Production Research*, *40*(5), 1193–1219. DOI: 10.1080/00207540110098823

Soomro, M. A., & Khan, A. N. (2024). Reimagining resilience: Visionary leadership, digital transformation, and strategic flexibility in small and medium enterprises in construction sector. *IEEE Transactions on Engineering Management*, *71*, 15070–15083. DOI: 10.1109/TEM.2024.3477629

Spohrer, J., Maglio, P. P., Bailey, J., & Gruhl, D. (2007). Steps toward a science of service systems. *Computer*, *40*(1), 71–77. DOI: 10.1109/MC.2007.33

Stolze, H. J., Mollenkopf, D. A., & Flint, D. J. (2016). What is the right supply chain for your shopper? Exploring the shopper service ecosystem. *Journal of Business Logistics*, *37*(2), 185–199. DOI: 10.1111/jbl.12122

Teece, D. J. (2007b). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal*, *28*(13), 1319–1350. DOI: 10.1002/smj.640

Teece, D. J. (2009). *Dynamic capabilities and strategic management: Organizing for innovation and growth*. Oxford University Press.

Teece, D. J. (2018). Business models and dynamic capabilities. *Long Range Planning*, *51*(1), 40–49. DOI: 10.1016/j.lrp.2017.06.007

Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, *18*(7), 509–533. DOI: 10.1002/(SICI)1097-0266(199708)18:7<509::AID-SMJ882>3.0.CO;2-Z

The Economist Intelligence Unit. (2023). *The cost-plus world of supply chains: The macroeconomic and geopolitical environment*. Economist Impact. https://impact.economist.com/projects/next-gen-supply-chains/images/articles/2023/report-1-report.pdf

Wieland, A. (2021). Dancing the supply chain: Toward transformative supply chain management. *The Journal of Supply Chain Management*, *57*(1), 58–73. DOI: 10.1111/jscm.12248

Wieland, A., & Durach, C. F. (2021). Two perspectives on supply chain resilience. *Journal of Business Logistics*, *42*(3), 315–322. DOI: 10.1111/jbl.12271

Wieland, A., & Wallenburg, C. M. (2013). The influence of relational competencies on supply chain resilience: A relational view. *International Journal of Physical Distribution & Logistics Management*, *43*(4), 300–320. DOI: 10.1108/IJPDLM-08-2012-0243

Wu, H., Li, G., & Zheng, H. (2024). How does digital intelligence technology enhance supply chain resilience? Sustainable framework and agenda. *Annals of Operations Research*, *355*(1), 901–923. DOI: 10.1007/s10479-024-06104-3

Yang, H., & Yu, W. (2025). Exploring the impacts of employee-based blockchain adoption on supply chain finance and operational performance. *International Journal of Production Research*, 1–19. DOI: 10.1080/00207543.2025.2580535

Dr. Yibin Li is a professor in the School of Economics and Management at Chang'an University. He holds a PhD in logistics engineering and management from Chang'an University. His research expertise includes smart warehousing, logistics platform operations management, and supply chain collaboration. He has supervised various research projects focusing on urban logistics, distribution network reengineering and logistics service chain governance. His academic work focuses on smart warehousing and supply chain resilience.

Aicha Ndiaye is a PhD candidate in logistics engineering and management at Chang'an University in Xi'an, China. She holds an MBA which informs her management-oriented approach to logistics research. Her research interests focus on supply chain resilience, the strategic integration of artificial intelligence, and human resource management within supply chain frameworks. Her work aims to bridge the gap between technological readiness and organizational resilience in global logistics networks. Her current work focuses on operational resilience and disruptions in road freight. She is dedicated to advancing scholarship through rigorous inquiry and interdisciplinary collaboration.